

MAGIC MAKER'S

ISTN212 - DATABASES AND PROGRAMMING

GROUP ASSIGNMENT – 2025

MASALA MAGIC™



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Project Background

Masala Magic is a small business that specialises in producing and selling food items such as Samosa's, Burfee, and other traditional snacks. Initially, the business operated in a local market where all transactions were carried out manually, with customers paying in cash and sellers calculating totals and change mentally. While this approach worked in a small-scale setting, it created several challenges, including human errors in calculations that led to customer dissatisfaction and revenue losses, unreliable stock tracking that made inventory management difficult, and the absence of proper sales records, which limited the ability to analyse performance and plan for growth.

Furthermore, because prices were not standardised or clearly displayed, customers often bargained with sellers, leading to inconsistent pricing and, in some cases, items being sold below their intended value, which reduced revenue and made it harder to track profits accurately.

Problem Description

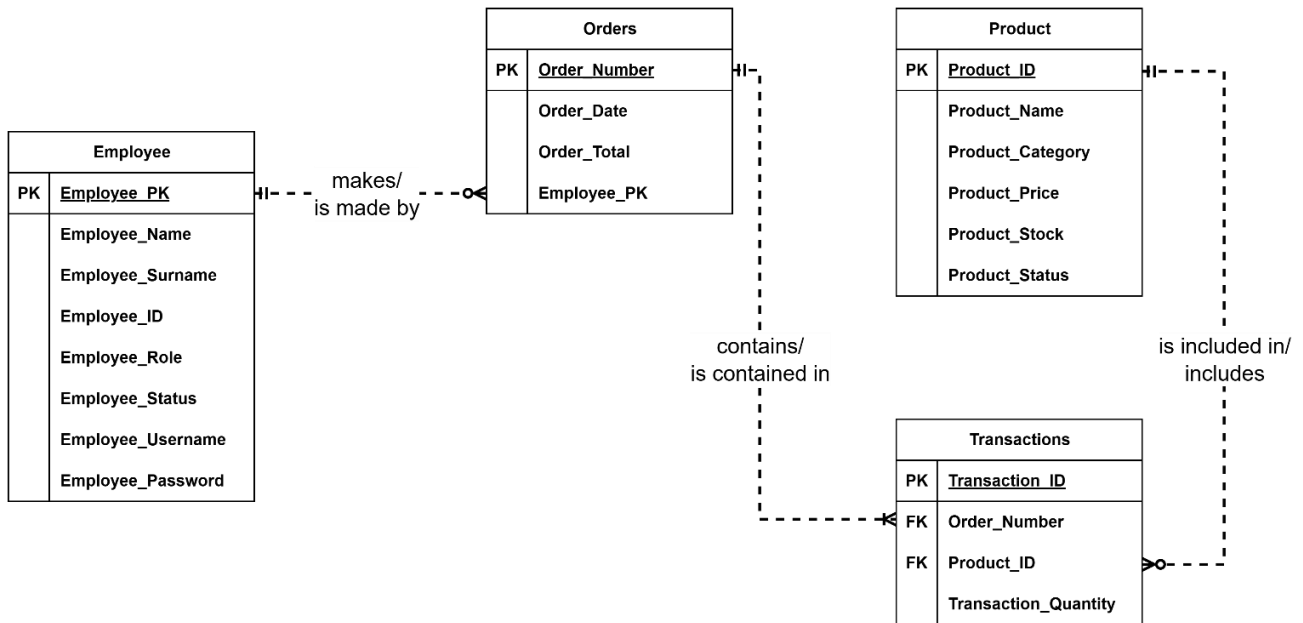
Recently, Masala Magic has expanded by moving into a permanent building to support larger operations, leading to increased transactions and higher customer demand. However, manual sales and stock tracking are prone to human error, time-consuming, and fail to provide reliable data for decision-making. Inconsistent pricing has also led to lost revenue and difficulties in maintaining profitability. Without a technology-driven solution, these issues will continue to negatively impact customer service, financial control, and the company's ability to grow.

Application Type

To address these challenges, Masala Magic has contracted our development team, **Magic Makers**, to design and implement a Point of Sale (POS) system. The proposed solution includes a digital cashier interface to streamline transactions and manage stock. This application is appropriate because it directly solves the identified problems by:

- Processing sales through a digital system.
- Supporting both cash and card payments.
- Automatically calculating totals and change to eliminate errors.
- Offering real-time stock and inventory management.
- Providing sales records and reports for management decisions.

ERD DESIGN



1. Employee – Orders Relationship

- An employee may process zero, one, or many orders.
- Each order must be processed by exactly one employee.

2. Orders – Product Relationship

- An order can contain one or more products.
- A product may appear in one or more orders.
- Since this is a many-to-many relationship, it is resolved using the Transactions table.

3. Transactions as a Junction Table

- Each transaction record is contained in exactly one order.
- Each transaction record includes exactly one product.
- A transaction must specify the quantity of the product in that order.
- An order must contain at least one transaction to be valid.

4. Orders – Transactions Relationship

- One order can have many transactions (one per product included in that order).
- Each transaction belongs to one and only one order.

5. Product – Transactions Relationship

- One product may appear in many transactions (if it is sold in different orders).
- Each transaction includes one and only one product.

SQL QUERIES AND EXPLANATIONS

SQL QUERIES	USEFULNESS
A. Basic Query SELECT * FROM Product;	- Shows all items sold at Masala Magic (e.g., Samoosa's, Burfee), including their prices, categories & stock count. - Management can check that all products are entered correctly in the database.
B. Update Query UPDATE Products SET Price = 7.00 WHERE ProductName = 'Chicken Samoosa';	- Updates the selling price when costs change. - Ensure customers are charged correctly and profit margins are maintained.
C. WHERE Clause Query (Daily Sales) SELECT * FROM Orders WHERE Order Date = '2025-07-04'; ('2025-07-04' can be replaced with other dates)	- Useful for reviewing daily sales. - Identify popular products sold on a given day (maybe during events, festivals, or promotions). - Quickly pull up all transactions for a given day to resolve customer complaints or refunds.
WHERE Clause Query (Low Stock) SELECT Product_Name, Product_Stock FROM Product WHERE Product_Stock < 15;	- It helps avoid stockouts by identifying items that need to be restocked. - Helps businesses time their supplier orders efficiently and negotiate better replenishment schedules.
WHERE Clause Query (Active Employees) SELECT Employee_Name, Employee_Surname, Employee_Role FROM Employee WHERE Employee_Status = 1;	Can be used to assign work schedules while preventing mistakes such as giving system access, salaries, or shifts to employees who no longer work there.

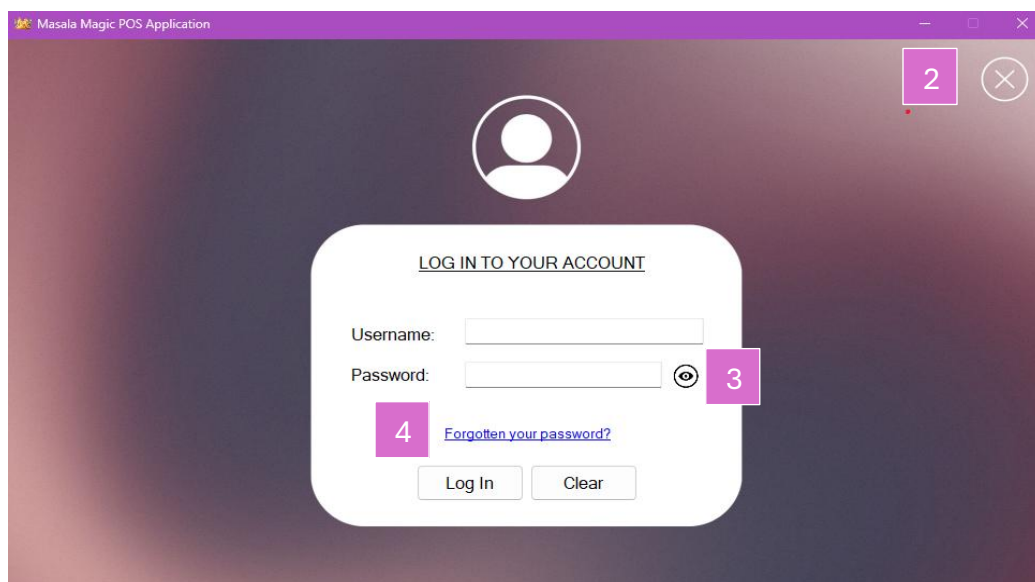
D. Subquery (Never Sold Products) <pre>SELECT Product_Name FROM Product WHERE Product_ID NOT IN (SELECT Product_ID FROM Transactions);</pre>	- Identifies products that have never been sold, allowing the business to plan discounts or promotions to clear stock. - Highlights risk for perishable food items, which may expire or go off if they remain unsold, causing financial loss and waste.
Subquery (Best-selling Product) <pre>SELECT Product_Name FROM Product WHERE Product_ID = (SELECT TOP 1 Product_ID FROM Transactions GROUP BY Product_ID ORDER BY SUM(Transaction_Quantity) DESC);</pre>	- Quickly identifies the top-selling product in your inventory. - Helps the business focus on marketing, promotions, and stock management for popular items.
E. Aggregation Query (Total Revenue) <pre>SELECT SUM(Transaction_Quantity * Product_Price) AS Total_Revenue FROM Transactions JOIN Product ON Transactions.Product_ID = Product.Product_ID;</pre>	- Provides insight into overall financial performance. - Helps management track sales trends and profitability.
F. Aggregation Query (Products Sold per Category) <pre>SELECT P.Product_Category, SUM(T.Transaction_Quantity) AS 'Total Sold' FROM Transactions T JOIN Product P ON T.Product_ID = P.Product_ID GROUP BY P.Product_Category;</pre>	- Helps identify the most and least popular product categories. - Guides the business in deciding which categories to expand or reduce.

USER INTERFACES



1. The progress bar at the bottom visually indicates the loading status of the application. As the system initializes resources such as connecting to the database, loading UI components, and preparing services, the green bar fills from left to right.

It provides feedback to the user, letting them know that the system is working and preventing confusion or impatience during the startup phase



2. Provides the user with the option to exit the application in a controlled and user-friendly manner.
3. Enables users to toggle the visibility of their password by clicking on a PictureBox containing “show” and “hide” eye icons, enhancing usability during login.

4. A clickable label that redirects users to a password reset form, allowing them to recover access if they have forgotten their credentials.

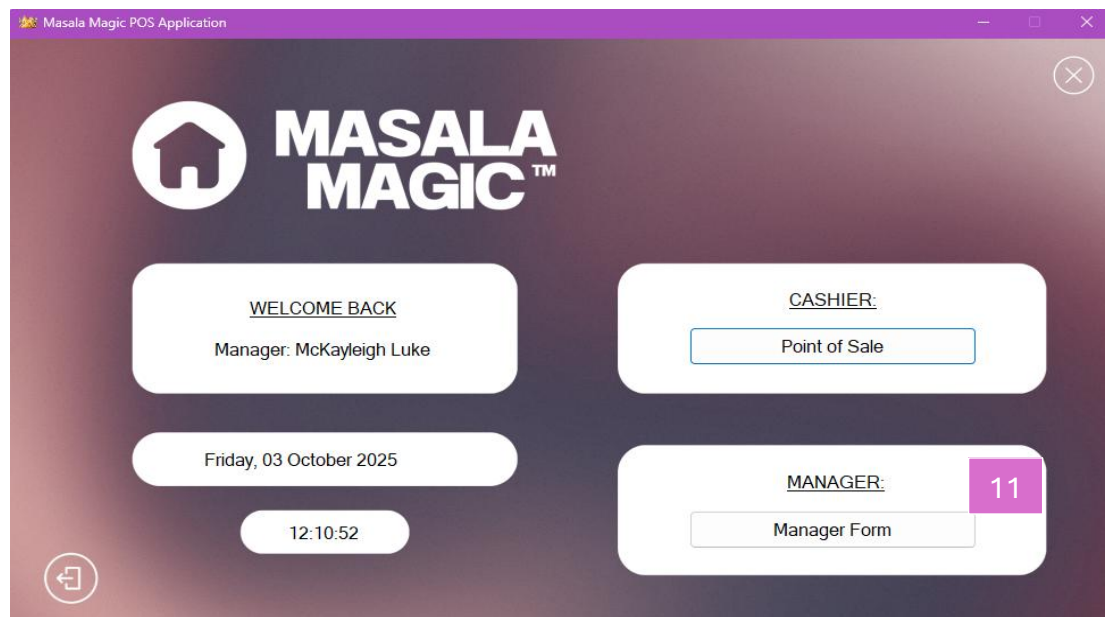
The screenshot shows the 'Masala Magic POS Application' window. At the top, there is a purple header bar. Below it, a large white padlock icon with a circular arrow around it is centered. Below the icon is a white rounded rectangle containing the text 'RESET YOUR PASSWORD'. Inside this rectangle, there are four input fields: 'Employee Username:', 'New Password:', 'Manager Username:', and 'Manager Password:'. The 'New Password' and 'Manager Password' fields have eye icons to toggle visibility. Below these fields are two buttons: 'Reset' and 'Clear'. A pink square with the number '5' is placed to the left of the 'Manager Username' and 'Manager Password' fields. In the bottom left corner of the application window, there is a back arrow icon and a pink square with the number '6'.

5. Employees can only update their credentials once a manager's username and password have been entered, ensuring secure and authorised access to sensitive account information.
6. Provides users with the option to return to the login form if they accidentally selected the "Forgot Password" feature.

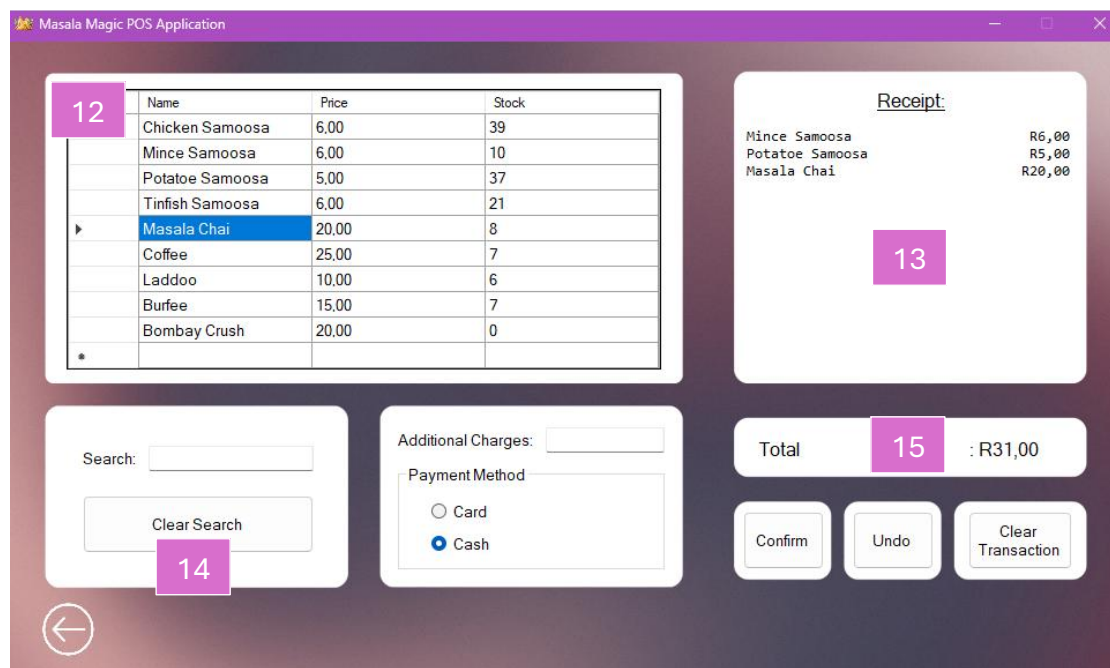
The screenshot shows the 'Masala Magic POS Application' window. At the top, there is a purple header bar. Below it, the 'MASALA MAGIC' logo is displayed. Below the logo, there is a white rounded rectangle containing the text 'WELCOME BACK' and 'Cashier: Keegan Pillay'. Below this, there is a white rounded rectangle containing the date 'Saturday, 04 October 2025'. Below the date, there is a white rounded rectangle containing the time '13:49:49'. To the right of these elements, there are two white rounded rectangles. The top one is labeled 'CASHIER:' and contains a button labeled 'Point of Sale'. The bottom one is labeled 'MANAGER:' and contains a button labeled 'Manager Form'. A pink square with the number '8' is placed to the left of the 'Point of Sale' button, and a pink square with the number '9' is placed to the left of the 'Manager Form' button. In the bottom left corner of the application window, there is a back arrow icon and a pink square with the number '10'.

7. Displays a personalised welcome message with the current date and time once the cashier logs in, creating a user-friendly and professional interface.
8. Allows the cashier to proceed to the POS form

9. Automatically disables the manager form button for cashier accounts, preventing unauthorised access to managerial functions.
10. Provides users with the option to securely log out of the system

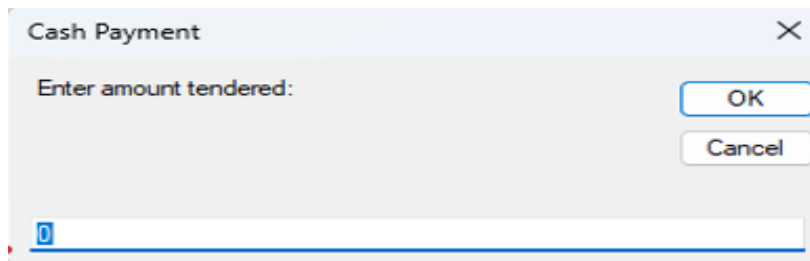


11. Once a manager logs into the system, the manager form is enabled and grants access to the necessary administrative functions and tasks.



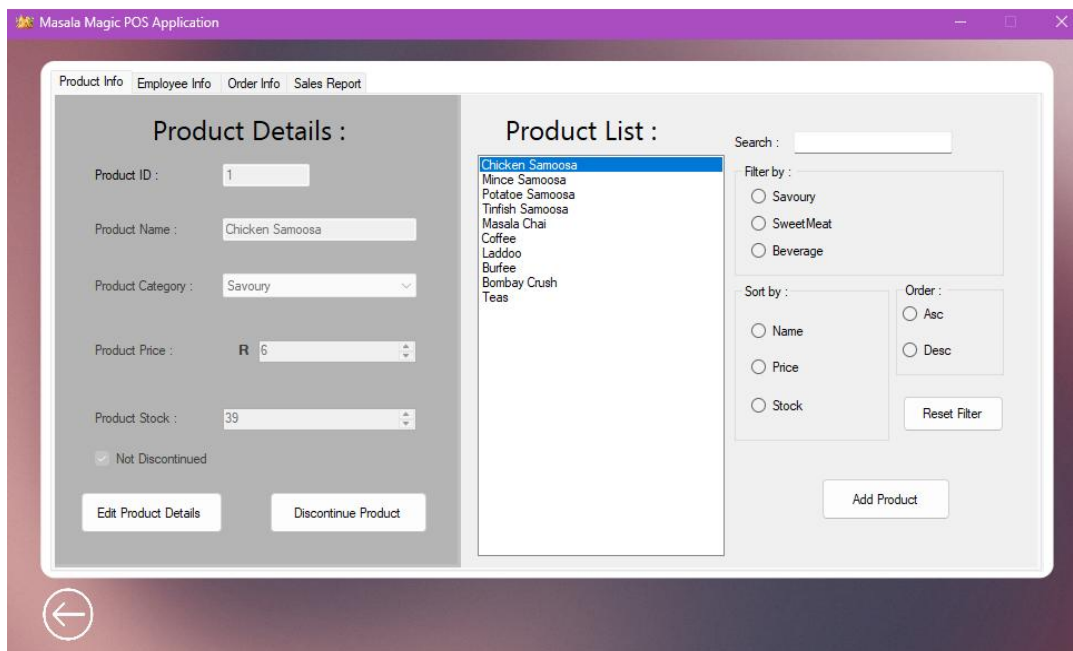
When the “Point of Sale” button is clicked, the system navigates to the above POS form, where transactions can be processed and managed.

12. A DataGrid that allows the cashier/manager to select products to add to the current order.
13. A multiline textbox (Receipt area) that displays the selected products along with their prices.
14. A search function that enables the user to quickly find specific products from the list.
15. A total display section that calculates and shows the total amount for the current transaction, including any additional charges added by the cashier.



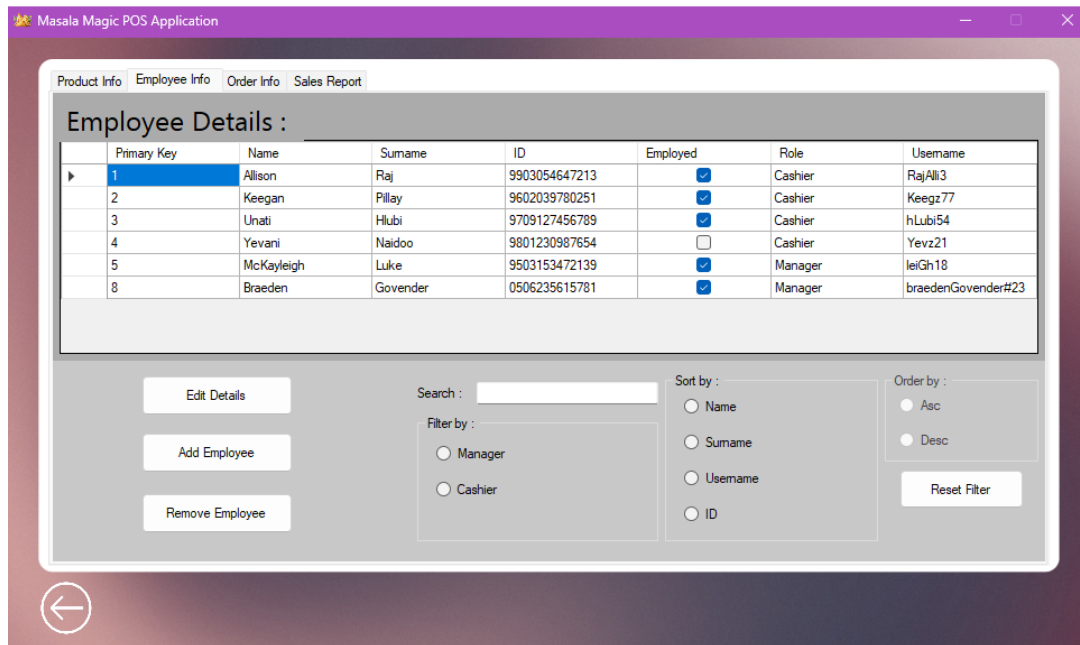
A dialog box titled "Cash Payment" with a close button (X) in the top right corner. It contains a label "Enter amount tendered:" followed by a text input field with a blue border and a numeric keypad icon on the left. To the right of the input field are two buttons: "OK" and "Cancel".

When the user selects Cash and clicks Confirm, an input box appears prompting them to enter the amount of cash received from the customer. The system then automatically calculates and displays the change due.

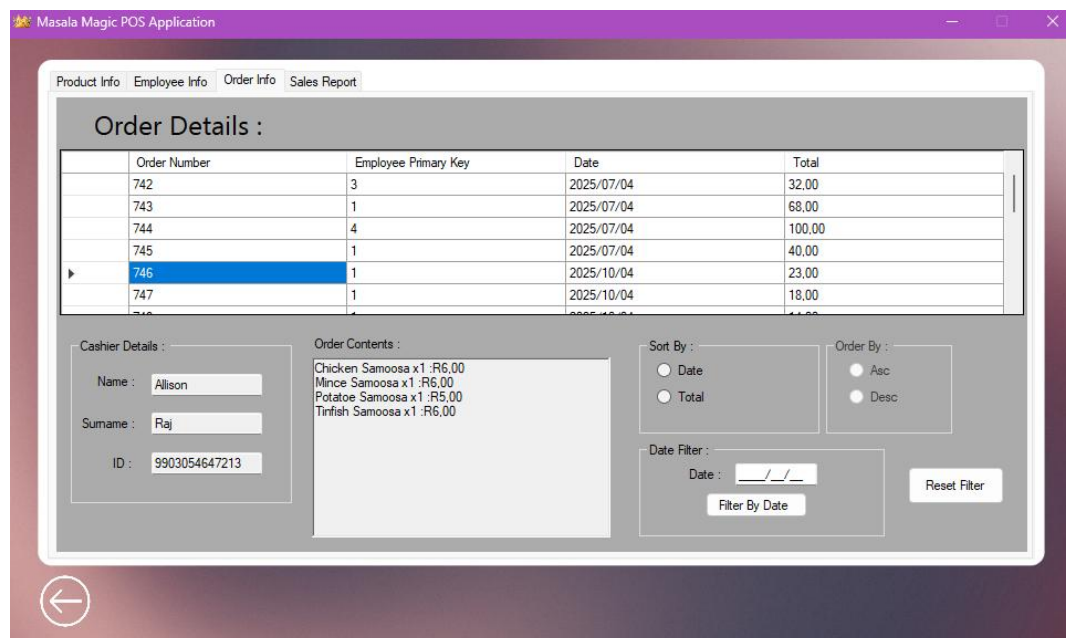


The screenshot shows the "Masala Magic POS Application" window. The "Product Info" tab is selected, displaying a "Product Details" form and a "Product List". The "Product Details" form includes fields for Product ID (1), Product Name (Chicken Samosa), Product Category (Savoury), Product Price (R 6), Product Stock (39), and a checkbox for "Not Discontinued". There are "Edit Product Details" and "Discontinue Product" buttons. The "Product List" shows a list of products: Chicken Samosa, Mince Samosa, Potatoe Samosa, Tinfish Samosa, Masala Chai, Coffee, Laddoo, Burfee, Bombay Crush, and Teas. To the right of the list are search and filter options, including a search bar, filter by category (Savoury, SweetMeat, Beverage), sort by (Name, Price, Stock), and order (Asc, Desc). There is a "Reset Filter" button and an "Add Product" button at the bottom right.

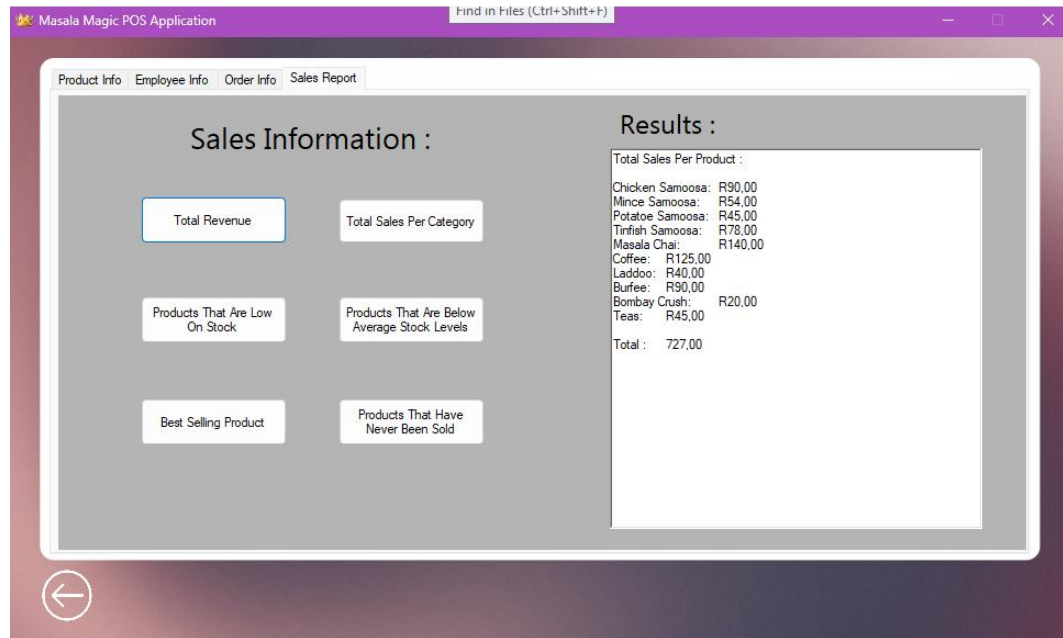
The Product Information Tab allows the manager to add, view, update, and delete product details, as well as search for specific products and filter items by different categories for easier management and navigation.



The Employee Information Tab allows the manager to efficiently manage employee records by adding, viewing, editing, or removing employee details as needed. It also provides search and filtering capabilities, making it easy to quickly locate specific employees.



The Order Details Tab allows the manager to view comprehensive information for a specific order, including which cashier processed it and the items it contains. Additionally, orders can be sorted by date or total amount, enabling efficient tracking and analysis of sales.



The Sales Information Tab provides the manager with a comprehensive overview of the business's sales performance. It allows the manager to view total revenue, track sales by product or category, and identify best-selling items. Additionally, it highlights products that are low on stock, below average stock levels, or have never been sold, helping with inventory management and restocking decisions.

IMPORTANT FEATURES INCLUDED IN THE APPLICATION

- **Login Facility:** A secure login system ensures only authorised personnel, like managers and cashiers, can access the application. This protects sensitive data and restricts access based on roles.
- **Conditional statements:** Used throughout the program to ensure certain code executes only when specific conditions are met.
- **Exception Handling:**
 - Handles unexpected errors, such as database connection failures or other runtime exceptions, by using **try-catch** statements to prevent the application from crashing and ensure smooth operation.
- **Data Validation:**
 - The system ensures users enter the correct data types and required information before saving to the database.
 - Numeric Validation – Ensures only numbers are entered in numeric fields.
 - Required Fields – Prevents users from leaving important fields empty.
 - Length Validation – Ensures input is within a specific length.
- **CRUD:** The system includes at least one of each CRUD operation (Create, Read, Update, Delete). The manager is allowed to perform these operations to efficiently manage records such as products, orders, and employees.
- **Search Facility:** Users can quickly find specific records, such as orders or products, using the search feature, which improves efficiency and reduces manual lookup time.
- **System Outputs:** Error Messages, Validation Messages, Warnings. The system provides clear feedback to users through message boxes, helping users correct mistakes, confirm actions, or respond to alerts.

FEATURES COVERED OUTSIDE OF THE CONTENT

- **Timer and Progress Bar:** A timer component is added to the welcome screen and configured with a specific interval (100 milliseconds), which determines how often the timer “ticks.” Each time the timer ticks, the progress bar’s value increases gradually, creating a smooth visual effect of the bar filling up over time.
- **.Start()** : The progress bar uses this method to begin filling automatically over a set interval defined by the timer. As the timer ticks, the progress bar gradually updates until it reaches 100%, signalling that the application is ready for use.
- **Different Types of Cursors:** The application enhances usability by changing the cursor icon depending on the action. For example, the pointer hand appears when hovering over buttons, providing a visual cue that the element is clickable.
- **.Pop()** : Allows the cashier to remove the last item added to the sales list, providing an easy way to correct mistakes. The items added to the sales list are managed using a stack-like structure, where the last item added is the first one removed when. pop() is used.
- **Input Boxes:** Prompts users to enter data dynamically, it was used to capture the cash amount provided by the customer, allowing the program to automatically calculate and display the correct change.
- **Dictionaries:** Used in the POS forms to efficiently track employees and stock numbers. It stores data as key-value pairs. Each key is unique and maps to a specific value, allowing fast lookup, addition, and updates of information such as employee IDs or product stock counts.

AI USE

AI tools (ChatGPT, DeepSeek) were used during the development and documentation of this project. Specifically, AI was used to:

- Help improve the structure and layout of written content for a professional finish.
- Enhance overall code cleanliness, organisation, and readability.
- Provide debugging assistance to help identify and explain errors in the code and suggested possible fixes.
- Suggest methods to test and validate SQL queries to ensure they returned the correct data.
- Explain unfamiliar programming concepts, tools, and components, which supported better understanding and correct application in the system.

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